Subject: Mathematics | **WEEKLY CHAPTER TEST – 03** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 6: Triangles** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Marks are indicated against each question.
2. **Section A** consists of 5 questions of 1 mark each (4 MCQs and 1 Assertion-Reason).
3. **Section B** consists of 4 questions of 11 marks total (1 VSAQ of 2 marks and 3 SA of 3 marks each).
4. **Section C** consists of 1 Case-Based integrated unit of assessment of 4 marks with sub-parts.
5. Use of calculators is strictly prohibited. Neat geometric sketches must accompany geometric proofs.

SECTION A: MULTIPLE CHOICE QUESTIONS (1 Mark Each)

1. If $\Delta ABC \sim \Delta DEF$ such that $2AB = DE$ and $BC = 8$ cm, then EF is equal to:

(a) 12 cm (b) 16 cm (c) 4 cm (d) 8 cm

2. A vertical stick 20 cm long casts a shadow 10 cm long on the ground. At the same time, a mobile tower casts a shadow 50 m long. The height of the tower is:

(a) 100 m (b) 120 m (c) 25 m (d) 200 m

3. It is given that $\Delta ABC \sim \Delta DFE$, $\angle A = 30°$, $\angle C = 50°$, $AB = 5$ cm, $AC = 8$ cm, and $DF = 7.5$ cm. Then which of the following is true?

(a) $DE = 12$ cm, $\angle F = 50°$ (b) $DE = 12$ cm, $\angle F = 100°$ (c) $EF = 12$ cm, $\angle D = 100°$ (d) $EF = 12$ cm, $\angle D = 30°$

4. The diagonals of a rhombus are 16 cm and 12 cm. The length of the side of the rhombus is:

(a) 9 cm (b) 10 cm (c) 8 cm (d) 20 cm

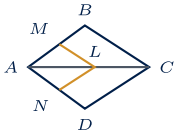
5. **Assertion (A):** If a line is drawn parallel to one side of a triangle intersecting the other two sides, then it divides the two sides in the same ratio.
Reason (R): This result is known as Basic Proportionality Theorem (Thales' Theorem).

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

SECTION B: DESCRIPTIVE QUESTIONS (PART 1)

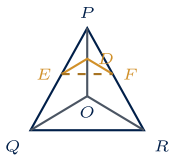
6. **[2 Marks]** In ΔABC, D and E are points on the sides AB and AC respectively. If $AB = 12$ cm, $AD = 8$ cm, $AE = 12$ cm, and $AC = 18$ cm, show whether $DE \parallel BC$.

7. **[3 Marks]** In the given figure, if $LM \parallel CB$ and $LN \parallel CD$, prove mathematically that:



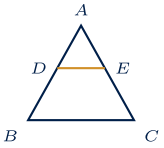
$$\frac{AM}{AB} = \frac{AN}{AD}$$

SECTION B: DESCRIPTIVE QUESTIONS (PART 2 Continued)



8. **[3 Marks]** In the given figure, $DE \parallel OQ$ and $DF \parallel OR$ in ΔPQR. Prove mathematically that:

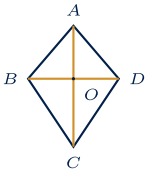
$$EF \parallel QR$$



9. **[3 Marks]** In ΔABC, D and E are points on sides AB and AC such that $DE \parallel BC$ and $BD = CE$. Prove that: (i) ΔADE is an isosceles triangle, (ii) ΔABC is an isosceles triangle.

SECTION C: CASE STUDY BASED QUESTION (4 Marks)

10. **Case Study 03: Aeronautical Kite Design & Geometric Proportions**



Rahul is designing a traditional diamond kite for an aerodynamics competition. The frame consists of two perpendicular bamboo sticks AC and BD intersecting at O, where AC is the vertical spine and BD is the horizontal crosspiece. In the upper half, $AB = AD = 25$ cm and $OA = 20$ cm.

(i) **[1 Mark]** Find the exact length of line segment OB.

(ii) **[1 Mark]** Which geometric similarity criterion proves that $\Delta AOB \sim \Delta AOD$?

(iii) **[2 Marks]** If a smaller replica is built with scale factor $2 : 5$, find the lengths of the crosspiece and spine of the miniature kite if original $AC = 45$ cm and $BD = 30$ cm.

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